Matt Maciejewski

Curriculum Vitae

✉ matt@mmaciejewski.com

Research Specialization

speech & audio processing, signal processing, deep learning, statistical modeling

Education

2014–2021 **Ph.D., Electrical Engineering**, *Johns Hopkins University*, Baltimore, MD.
Advisors: Sanjeev Khudanpur and Shinji Watanabe
Dissertation Topic: speech separation in noisy and reverberant conditions

2014–2017 **M.S., Electrical and Computer Engineering**, *Johns Hopkins University*, Baltimore, MD.

2010–2014 **B.S., Electrical and Computer Engineering**, *Carnegie Mellon University*, Pittsburgh, PA.

Experience

2023–Present **Associate Research Scientist**, *Human Language Technologies Center of Excellence, Johns Hopkins University*, Baltimore, MD.
Research in speech technologies focused on audio captured by a distant microphone, including speech enhancement and speaker diarization.

2021–2023 **Research Scientist**, *Amazon Alexa AI*, Cambridge, MA.
Research in acoustic event detection and acoustic scene classification. Leveraged knowledge distillation and data mining to deploy improved event detectors with a lower on-device resource footprint. Mentored interns that led to return offers.

2014–2021 **Research Assistant**, *Department of Electrical and Computer Engineering, Johns Hopkins University*, Baltimore, MD.
Research in speech processing with emphasis on conversational speech, primarily far-field speech separation. This included speaker diarization, speaker recognition, and speech enhancement, and contributions to the Kaldi ASR toolkit.

2019–2019 **Research Intern**, *Mitsubishi Electric Research Laboratories (MERL)*, Cambridge, MA.
Worked on noisy and reverberant speech separation, releasing the WHAMR! corpus and improved performance through cascaded model architecture.

2013–2014 **Undergraduate Researcher**, *Carnegie Mellon University*, Pittsburgh, PA.
Worked on dereverberation of speech signals in binaural recordings.

Professional Activities

Organizer CHiME-8, CHiME-7

Session Chair Interspeech 2025, Odyssey 2024

Reviewer	ICASSP, Interspeech, SLT, ASRU, DCASE, WASPAA, IWAENC, CHiME, IEEE Open Journal of Signal Processing, Applied Acoustics, Computer Speech and Language, Speech Communication

Teaching Experience

Co-Instructor	520.666 Information Extraction. *JHU* 2026
Guest Lecturer	601.667 Introduction to Human Language Technology. *JHU* 2023–Present
	JSALT 2024 Summer School

Student Development

PhD Co-Advisor	Xavier Velez. *Audio Representation Learning*
Qualifying Project	Xavier Velez. *Neural Audio Codec Dereverberation*
	Xiluo He. *Stream-Conditioned Multi-Talker ASR*. Published at ICASSP 2026
Masters Thesis	Dominik Klement. *Unsupervised Speech Enhancement*. Masters Thesis Award
Research Project	David Palzer
PhD Committee	Yiwen Shao
GBO Committee	Neha Verma, Zili Huang

Grants and Funding

Under review	NSF Future CoRe *Native Phase Processing via Complex-Valued Neural Networks for Speech Enhancement and Acoustic Beamforming*

Awards

CHiME-8	Jury Prize for Task 2: NOTSOFAR

Publications

2027 S. Cornell, **M. Maciejewski**. *SNOAR: Self-Training with Noise Filtering, Observation Adding, and Remixing for Speech Denoising without Clean References*. (under review)

– X. Velez, **M. Maciejewski**, S. Khudanpur. *Towards Integrating Dereverberation Into A Neural Audio Codec-Based Processing Chain*. (under review)

– R. Kriz, D. Etter, A. Martin, C. Carpenter, D. Chakraborty, H. Recknor, R. Iranmanesh, **M. Maciejewski**, K. Murray, E. Yang, B. Van Durme, A. S. White, A. Yates, W. Walden. *MultiVENT-Raw: A Benchmark for Retrieval and Reasoning over Raw Videos*. (under review)

2026 **M. Maciejewski**, S. Cornell. *Ring Mixing with Auxiliary Signal-to-Consistency-Error Ratio Loss for Unsupervised Denoising in Speech Separation*. IEEE SLT. 2026

– D. Klement, **M. Maciejewski**, S. Khudanpur, J. Černocký, L. Burget. *Analysing Adversarial Priors for Data-driven Unsupervised Speech Enhancement*. ISCA Interspeech. 2026

– **M. Maciejewski**, S. Cornell. *Exploiting Noise Inseparability for Weakly-Supervised Discriminative Speech Denoising Using Noisy Targets*. IEEE IWAENC. 2026

– **M. Maciejewski**. *Single-Microphone Audio Point Source Discriminative Localization from Reverberation Late Tail Estimation*. IEEE ICASSP. 2026

- X. He, A. Polok, J. Villalba, T. Thebaud, **M. Maciejewski**. *Scaling Multi-Talker ASR with Speaker-Agnostic Activity Streams*. IEEE ICASSP. 2026
- Z. Huang, **M. Maciejewski**, L.P. Garcia Perera, S. Watanabe, S. Khudanpur. *CSPB: Conversational Speech Processing Benchmark for Self-supervised Speech Models*. EACL. 2026

2025 **M. Maciejewski**. *Speaker Conditioning of Voice Activity Detection via Implicit Separation*. ISCA Interspeech. 2025

- D. Palzer, **M. Maciejewski**, E. Fosler-Lussier. *End-to-End Diarization utilizing Attractor Deep Clustering*. ISCA Interspeech. 2025

2024 A. Polok, D. Klement, J. Han, Š. Sedláček, B. Yusuf, **M. Maciejewski**, M. Wiesner, L. Burget. *BUT/JHU System Description for CHiME-8 NOTSOFAR-1 Challenge*. Proc. CHiME-8. 2024

- S. Cornell, T.J. Park, H. Huang, C. Boeddeker, X. Chang, **M. Maciejewski**, M. Wiesner, P. Garcia, S. Watanabe. *The CHiME-8 DASR Challenge for Generalizable and Array Agnostic Distant Automatic Speech Recognition and Diarization*. Proc. CHiME-8. 2024
- **M. Maciejewski**, D. Klement, R. Huang, M. Wiesner, S. Khudanpur. *Evaluating the Santa Barbara Corpus: Challenges of the Breadth of Conversational Spoken Language*. ISCA Interspeech. 2024
- D. Raj, M. Wiesner, **M. Maciejewski**, L.P. Garcia-Perera, D. Povey, S. Khudanpur. *On Speaker Attribution with SURT*. ISCA Odyssey. 2024
- D. Palzer, **M. Maciejewski**, E. Fosler-Lussier. *Improving Neural Diarization Through Speaker Attribute Attractors and Local Dependency Modeling*. IEEE ICASSP. 2024

2023 S. Cornell, M. Wiesner, S. Watanabe, D. Raj, X. Chang, P. Garcia, **M. Maciejewski**, Y. Masuyama, Z.-Q. Wang, S. Squartini, S. Khudanpur. *The CHiME-7 DASR Challenge: Distant Meeting Transcription with Multiple Devices in Diverse Scenarios*. Proc. CHiME-7. 2023

- **M. Maciejewski**, J. Shi, S. Watanabe, S. Khudanpur. *A Dilemma of Ground Truth in Noisy Speech Separation and an Approach to Lessen the Impact of Imperfect Training Data*. Computer Speech & Language. 2023

2021 **M. Maciejewski**. *Towards Single-Channel Speech Separation in Noise and Reverberation*. [Doctoral dissertation]. Johns Hopkins University. 2021

- **M. Maciejewski**, S. Watanabe, S. Khudanpur. *Speaker Verification-Based Evaluation of Single-Channel Speech Separation*. ISCA Interspeech. 2021
- **M. Maciejewski**, J. Shi, S. Watanabe, S. Khudanpur. *Training Noisy Single-Channel Speech Separation with Noisy Oracle Sources: A Large Gap and a Small Step*. IEEE ICASSP. 2021

2020 **M. Maciejewski**, G. Wichern, E. McQuinn, J. Le Roux. *WHAMR!: Noisy and Reverberant Single-Channel Speech Separation*. IEEE ICASSP. 2020

- A. Arora, D. Raj, A.S. Subramanian, K. Li, B. Ben-Yair, **M. Maciejewski**, P. Żelasko, P. Garcia, S. Watanabe, S. Khudanpur. *The JHU Multi-Microphone Multi-Speaker ASR System for the CHiME-6 Challenge*. Proc. CHiME-6. 2020

2019 **M. Maciejewski**, G. Sell, Y. Fujita, L.P. Garcia-Perera, S. Watanabe, and S. Khudanpur. *Analysis of Robustness of Deep Single-Channel Speech Separation Using Corpora Constructed From Multiple Domains*. IEEE WASPAA. 2019

2018 N. Kanda, R. Ikeshita, S. Horiguchi, Y. Fujita, K. Nagamatsu, X. Wang, V. Manohar, N.E.Y. Soplin, **M. Maciejewski**, S.J. Chen, A.S. Subramanian, R. Li, Z. Wang, J. Naradowsky, L.P. Garcia-Perera, G. Sell. *The Hitachi/JHU CHiME-5 system: Advances in speech recognition for everyday home environments using multiple microphone arrays*. Proc. CHiME-5. 2018

– G. Sell, D. Snyder, A. McCree, D. Garcia-Romero, J. Villalba, **M. Maciejewski**, V. Manohar, N. Dehak, D. Povey, S. Watanabe, and S. Khudanpur. *Diarization is Hard: Some Experiences and Lessons Learned for the JHU Team in the Inaugural DIHARD Challenge*. ISCA Interspeech. 2018

– **M. Maciejewski**, D. Snyder, V. Manohar, N. Dehak, and S. Khudanpur. *Characterizing Performance of Speaker Diarization Systems on Far-field Speech using Standard Methods*. IEEE ICASSP. 2018

– N. Ryant, E. Bergelson, K. Church, A. Cristia, J. Du, S. Ganapathy, S. Khudanpur, D. Kowalski, M. Krishnamoorthy, R. Kulshreshta, M. Liberman, Y.D. Lu, **M. Maciejewski**, F. Metze, J. Profant, L. Sun, Y. Tsaok, Z. Yum. *Enhancement and Analysis of Conversational Speech: JSALT 2017*. IEEE ICASSP. 2018

2016 E. Chodroff, **M. Maciejewski**, J. Trmal, S. Khudanpur, J. Godfrey. *New release of Mixer-6: Improved validity for phonetic study of speaker variation and identification*. ELRA LREC. 2016

2015 H. Hermansky, L. Burget, J. Cohen, E. Dupoux, N. Feldman, J. Godfrey, S. Khudanpur, **M. Maciejewski**, S.H. Mallidi, A. Menon, T. Ogawa, V. Peddinti, R. Rose, R. Stern, M. Wiesner, K. Veselý. *Towards machines that know when they do not know: Summary of work done at 2014 Frederick Jelinek Memorial Workshop*. IEEE ICASSP. 2015

– H.M. Park, **M. Maciejewski**, C. Kim, R. Stern. *Robust automatic speech recognition in reverberation: onset enhancement versus binaural source separation*. The Journal of the Acoustical Society of America. 2015

2014 H.M. Park, **M. Maciejewski**, C. Kim, R. Stern. *Robust speech recognition in reverberant environments using subband-based steady-state monaural and binaural suppression*. ISCA Interspeech. 2014